

MPCA Module 2 — 2 Mark Questions & Answers

Q1. Define instruction format.

An **Instruction Format** defines the internal layout and binary structure of a machine instruction. It specifies the bit fields allocated for the **Opcode** (operation code) and **Operands** (registers, memory addresses, or immediate values).

Q2. What is a zero-address instruction format? Give an example.

A **Zero-Address Instruction Format** contains only an Opcode and specifies **no explicit address fields**. It operates implicitly on a **Stack** data structure where operands are pushed/popped from the top of stack (**TOS**).

* **Example:** `ADD` (Pops top two elements from stack, adds them, and pushes result onto stack).

Q3. Define control unit. What are its functions?

The **Control Unit (CU)** is the engine of the CPU that coordinates all computer hardware operations.

* **Functions:**

- 1. Fetches and decodes instructions from memory.
 - 2. Generates micro-operation timing and control signals to direct the ALU, registers, and memory bus.
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Q4. What is the difference between hardwired and microprogrammed control unit? (any 2 points)

Feature	Hardwired Control Unit	Microprogrammed Control Unit
Implementation	Fixed combinational logic gates and flip-flops.	Microprograms stored in Control Memory (ROM).
Flexibility	Rigid; modification requires rewiring hardware.	Flexible; modification done by updating microcode.

Q5. Define microinstruction.

A **Microinstruction** is a low-level control word stored in Control Memory (ROM) that specifies one or more primitive hardware micro-operations (e.g., register transfer, ALU operation) to be executed within a single clock cycle.

Q6. What is a pipeline hazard?

A **Pipeline Hazard** is a situation or conflict in an instruction pipeline that prevents the next instruction from executing in its designated clock cycle, introducing pipeline stalls (bubbles) and reducing instruction throughput.

Q7. Define Flynn's classification. Name its four types.

Flynn's Classification is a taxonomy for categorizing computer architectures based on the number of concurrent **Instruction Streams** and **Data Streams**.

* **Four Types:**

1. **SISD:** Single Instruction, Single Data
 2. **SIMD:** Single Instruction, Multiple Data
 3. **MISD:** Multiple Instruction, Single Data
 4. **MIMD:** Multiple Instruction, Multiple Data
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Q8. What is instruction pipelining?

Instruction Pipelining is an architectural technique where the execution of sequential instructions is overlapped in time across multiple independent processing stages (e.g., Fetch, Decode, Execute), increasing instruction throughput.

Q9. Define structural hazard.

A **Structural Hazard** (resource conflict) occurs in a pipeline when two or more overlapping instructions require access to the exact same physical hardware resource (e.g., single memory bus or ALU) in the same clock cycle.

Q10. What is branch prediction?

Branch Prediction is a technique used by modern processors to predict the outcome of a conditional branch instruction (taken or not taken) before it is actually evaluated, allowing the pipeline to prefetch instructions without stalling.

Q11. State the functions of PC, IR, MAR, and MDR.

- **PC (Program Counter):** Holds the memory address of the next instruction to be fetched.
 - **IR (Instruction Register):** Stores the current instruction byte while it is decoded.
 - **MAR (Memory Address Bus Register):** Holds the physical address sent to the memory bus.
 - **MDR (Memory Data Register):** Holds data read from or being written to main memory.
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Q12. What is RAW hazard?

A **RAW (Read-After-Write) Hazard** (also called True Data Dependency) occurs when an instruction tries to read a register or memory location before a preceding instruction has finished writing its updated result to it.
